TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

Reminders
Words: at least ≥ · at most ≤ · more than > · fewer than <.
Solve like an equation: undo + and −, then × and ÷.
Flip the sign when you multiply or divide both sides by a negative number.
Graph: ○ for < and >, ● for ≤ and ≥. Shade right for > and ≥, left for < and ≤.

1. Elijah starts with a $25 credit and earns $3.75 for each box he sells. Which equation gives his total sales, y, after selling x boxes?

Tip: [[y = mx + b]]: [[m]] = rate, [[b]] = start

A. $y = 25x + 3.75$

B. $y = 3.75x + 25$

C. $y = -3.75x + 25$

Start $25 ($b$), $3.75 per box ($m$). Like test Q1.

2. Which symbol means "at least"?

Tip: at least = this much or more

F. ≤

G. >

H. ≥

At least 5 means 5 or more: ≥.

3. Which symbol means "no more than"?

> **Tip:** no more than = this much or less

A. ≥

B. <

C. ≤

No more than 5 means 5 or less: ≤.

4. What value of b makes $3.75b + 25 = 400$ true?

> **Tip:** subtract the start, then divide by the rate

F. 100

G. 106.67

H. 113.33

Subtract 25: $3.75b = 375$. Divide: $b = 100$.